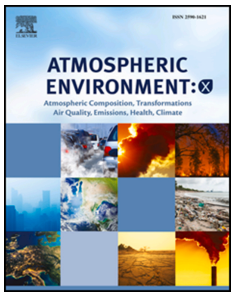
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Effects of air-conditioning operation and high ambient temperature on CO, NOx, and particle number emissions from heavy-duty diesel vehicles under CHTC cycles

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PII: S2590-1621(26)00122-X

DOI: https://doi.org/10.1016/j.aeaoa.2026.100534

Reference: AEAOA 100534

To appear in: *Atmospheric Environment: X*

Received Date: 13 July 2026

Revised Date: 26 October 2026

Accepted Date: 5 November 2026

Please cite this article as: Liu, Y., peng, D., Zu, L., Jiang, H., Zhang, H., Effects of air-conditioning operation and high ambient temperature on CO, NOx, and particle number emissions from heavy-duty diesel vehicles under CHTC cycles, *Atmospheric Environment: X*, https://doi.org/10.1016/j.aeaoa.2026.100534.

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Effects of air-conditioning operation and high ambient temperature on CO, NOx, and particle number emissions from heavy-duty diesel vehicles under CHTC cycles

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Abstract: Air-conditioning operation imposes an additional auxiliary load on heavy-duty diesel vehicles, yet its effects on conventional pollutant emissions, particularly under high ambient temperatures, remain insufficiently understood. In this study, six heavy-duty diesel vehicles, including dump trucks, freight trucks, a coach, and a tractor-trailer, were tested over vehicle-specific China Heavy-duty Commercial Vehicle Test Cycles in a climate-controlled chassis dynamometer laboratory. Experiments were conducted at ambient temperatures of 32 and 42 ℃ with air conditioning switched off and on. Carbon monoxide (CO), nitrogen oxides (NOx), and particle number (PN) emissions were evaluated using whole-cycle distance- and work-based emission factors, stage-resolved analysis, transient emission profiles, a high-temperature modulation index, and exploratory machine-learning analysis. At 32 ℃, air-conditioning operation produced pollutant and vehicle-specific responses. Whole-cycle CO emissions changed only moderately, whereas NOx responses varied in both direction and magnitude among vehicles. PN exhibited the most consistent positive response, with higher distance-based emission factors observed for all six vehicles during air-conditioning operation. Stage-resolved analysis showed that these effects were not uniformly distributed throughout the driving cycles: CO responses were concentrated mainly in

low-speed or urban stages, while NOx and PN responses depended on specific vehicle–stage combinations. Increasing the ambient temperature from 32 to 42 ℃ increased the AC-induced relative changes in distance-based NOx for Vehicle E and PN for Vehicle F by 123.7 and 82.7 percentage points, respectively, while the CO interaction indices were mostly negative. Exploratory machine-learning results further indicated scenario-dependent changes in the variables associated with second-by-second emission variability. Overall, the results show that air-conditioning-related emissions from heavy-duty diesel vehicles vary by pollutant, vehicle, driving stage, and ambient temperature, highlighting the need to consider auxiliary cooling loads when evaluating emissions under warm and high-temperature operating conditions.

Keywords: Heavy-duty diesel vehicles; Air-conditioning operation; High ambient temperature; Conventional pollutant emissions; Particle number emissions

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1. Introduction

Heavy-duty vehicles are essential for freight transport, construction activity, and intercity mobility, but they also make a disproportionate contribution to road-transport NOx and particulate emissions because of their high engine power, long annual mileage, and diverse duty cycles[1]. Advanced exhaust aftertreatment systems have substantially reduced emissions from modern diesel heavy-duty vehicles. Nevertheless, tailpipe emissions remain highly variable during transient operation and under changing thermal boundary conditions, particularly when engine load and aftertreatment temperature deviate from their optimal ranges[2-5]. Auxiliary loads that alter engine work and exhaust temperature may therefore become increasingly relevant to the real-world emission performance of modern heavy-duty vehicles.

Air-conditioning operation is a common auxiliary load under warm and hot

ambient conditions. Compressor operation increases engine power demand and may alter fuel consumption, combustion conditions, exhaust flow, and exhaust temperature[6]. The resulting tailpipe response is not necessarily uniform across pollutants. CO reflects the combined effects of engine-out formation and oxidation in the diesel oxidation catalyst; tailpipe NOx depends on both engine-out formation and selective catalytic reduction efficiency; and particle number (PN) emissions may be influenced by engine-out particle formation, diesel particulate filter status, and regeneration-related events[7]. In addition, high ambient temperature can increase cabin cooling demand and strengthen air-conditioning-related load, while also modifying engine thermal management, exhaust temperature and aftertreatment operating conditions [8-10]. Therefore, air-conditioning and ambient temperature may produce pollutant- and vehicle-specific emission responses rather than a uniform increase in all pollutants.

More recent studies have primarily focused on fuel consumption, CO_2 emissions, thermal comfort, and electric-vehicle driving range. Costa and Valle [11] compared a tetra-fuel vehicle under air-conditioning-on and air-conditioning-off conditions and found that the resulting increase in CO_2 emissions varied considerably across fuels and operating conditions. Using second-by-second real-world data, Mogro and Huertas [12] reported that the additional distance-based fuel consumption attributable to air-conditioning operation was greatest at vehicle speeds below 10 km h^{-1}, reaching 17.6 L per 100 km, whereas the additional hourly fuel consumption remained approximately 0.9 L/h. Their findings also demonstrated that the apparent air-conditioning penalty depends strongly on the metric used to quantify fuel consumption. For commercial vehicles, Zacharof et al. [13] investigated a city bus and showed that HVAC demand, environmental conditions, and passenger occupancy jointly affect vehicle energy consumption and on-road CO_2 emissions. Seo et al. [14] further identified HVAC energy demand as a major contributor to temperature-related driving-range losses in electric vehicles, while Kaddoura et al. [15] applied machine-learning-based correlation analysis to examine the relationship between air-conditioning use and vehicle fuel consumption. Collectively, these studies confirm the importance of air-conditioning as

an auxiliary energy consumer. However, their response variables have predominantly been limited to fuel use, energy consumption, CO_2 emissions, and electric-vehicle driving range, while direct evidence regarding the effects of air-conditioning operation on CO, NOx, and PN emissions remains limited.

Considerable progress has also been made in understanding the effects of driving conditions and aftertreatment thermal states on emissions from heavy-duty vehicles. Real-world studies of Euro VI and China 6 heavy-duty diesel vehicles have shown that NOx emissions can vary substantially with vehicle vocation, urban or highway operation, payload, exhaust temperature, and SCR activity[7, 16, 17]. McCaffery et al.[7] observed elevated brake-specific NOx emissions at low SCR operating temperatures, while Ge et al. associated the higher NOx emissions of urban vehicles with longer periods of low SCR temperature. Li et al. [18] demonstrated that SCR effectiveness, vehicle speed, and the air–fuel ratio jointly influence gaseous emissions from China V and China VI heavy-duty diesel vehicles. Regarding particulate emissions, Saari et al. [19] showed that high-load operation may increase both NOx and PN emission factors and that PN emissions can be strongly affected by nucleation-mode particles. Zheng et al.[3] further demonstrated that advanced aftertreatment configurations substantially influence the emission characteristics of particles larger than 10 nm. PEMS- and OBD-based studies have also established that vehicle speed, acceleration, engine load, road type, and vehicle category contribute to the pronounced variability in heavy-duty vehicle emissions [4, 5, 20]. Nevertheless, these studies have generally examined driving conditions or aftertreatment performance in isolation, without explicitly considering the additional air-conditioning load imposed under high-temperature conditions or its potential interactions with engine operation and aftertreatment thermal states.

This study investigated the effects of air-conditioning operation on CO, NOx, and PN emissions from six heavy-duty diesel vehicles tested over vehicle-specific China Heavy-duty Commercial Vehicle Test Cycles (CHTC). Experiments were conducted at two ambient temperatures (32 and 42 °C) with air conditioning switched off and on. Whole-cycle distance- and work-based emission factors were evaluated to quantify the

overall effects of air-conditioning operation, while stage-resolved and transient analyses were performed to identify the driving conditions associated with these responses. The influence of high ambient temperature on the direction and magnitude of the AC-related emission responses was also examined. Because vehicle mass, vocation, test cycle, engine characteristics, and aftertreatment configuration were not independently controlled, differences among the tested vehicles were interpreted descriptively as vehicle-specific responses.

2. Materials and Methods

2.1 Tested vehicles and experimental system

Six diesel heavy duty vehicles were tested in this study, covering four representative vehicle categories: dump trucks, freight trucks, a coach and a tractor trailer. To avoid identifying individual vehicles, they were denoted as Vehicles A-F. Vehicles A and D were dump trucks, Vehicles B and E were freight trucks, Vehicle C was a coach, and Vehicle F was a tractor trailer. The vehicle masses ranged from 4.5 t to 49 t, providing a broad range of vehicle sizes and transport applications. The corresponding vehicle information is summarized in Table 1.

Table 1. Main parameters of the tested heavy-duty vehicles.

Vehicle ID	A	B	C	D	E	F
Vehicle Type	Dump truck	Freight truck	Coach	Dump truck	Freight truck	Tractor trailer
Vehicle Mass (t)	4.5	7.3	12.9	25	31	49
Test Cycle	CHTC-D	CHTC-HT	CHTC-C	CHTC-D	CHTC-HT	CHTC-TT

The experiments were conducted in a climate-controlled chassis dynamometer laboratory. The chassis dynamometer was used to reproduce the prescribed China Heavy-duty Commercial Vehicle Test Cycle profiles, and the climate chamber was used to control the ambient temperature. The vehicle testing system is shown in Fig.1. The

system mainly consisted of the climate chamber, chassis dynamometer, wind resistance simulation system, vehicle air-conditioning system, fuel consumption meter, exhaust flow measurement system and portable emission measurement system. CO and NOx were measured using the gas analysis module, and PN was measured using the particle number module. The exhaust flow rate was measured simultaneously, allowing second by second pollutant emission rates to be calculated from concentration and exhaust flow data. Vehicle speed, acceleration, torque, engine work and fuel related signals were also synchronously recorded for subsequent emission factor calculation and machine learning analysis.

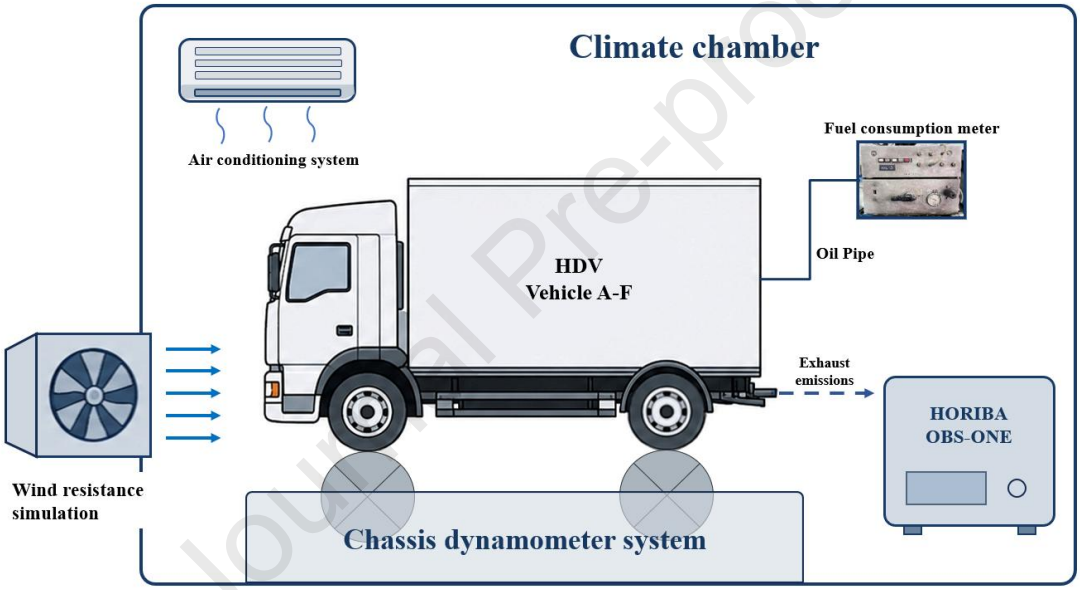


Fig.1. Schematic diagram of the heavy-duty vehicle testing system.

2.2 Test conditions and experimental design

The test was designed to evaluate the effects of ambient temperature and air-conditioning operation on CO, NOx and PN emissions. Two ambient temperatures were investigated: 32 ℃, defined as the reference temperature for this experimental design, and 42 ℃, defined as the high-temperature condition. The reference temperature represents a warm ambient condition and is not intended to denote a standard laboratory temperature. Under each temperature condition, tests were conducted with the vehicle air-conditioning system turned off and turned on. When air-conditioning was activated, the system was operated in cooling mode with maximum fan speed and recirculation

mode to maintain consistent boundary conditions. During all tests, windows, doors and other openings were kept closed.

For each vehicle, four temperature and air-conditioning scenarios were tested: 32℃ without air-conditioning, 32℃ with air-conditioning, 42℃ without air-conditioning and 42℃ with air-conditioning. Each scenario was repeated twice. Therefore, a total of 48 valid CHTC tests were conducted. The test matrix is summarized in Table 2. Whole-cycle and stage-resolved emission factors were summarized as arithmetic means of the two repeated tests for each vehicle and scenario. Error bars in Figs. 3-5 indicate ±1 sample standard deviation (SD; n = 2), reflecting variation between the repeated tests.

Table 2. Experimental design of temperature and air-conditioning scenarios.

Ambient temperature	Air-conditioning state	Scenario description	Repetition
32℃	Off	Reference temperature without air-conditioning	2
32℃	On	Reference temperature with air-conditioning	2
42℃	Off	High temperature without air-conditioning	2
42℃	On	High temperature with air-conditioning	2

Vehicle specific CHTC cycles were selected according to vehicle category. Vehicles A and D were tested under CHTC-D, Vehicles B and E were tested under CHTC-HT, Vehicle C was tested under CHTC-C, and Vehicle F was tested under CHTC-TT. And the cycles were further divided according to their speed characteristics. Vehicles A, D and F were divided into low-speed and high-speed stages, whereas Vehicles B, C and E were divided into urban, suburban and highway stages.

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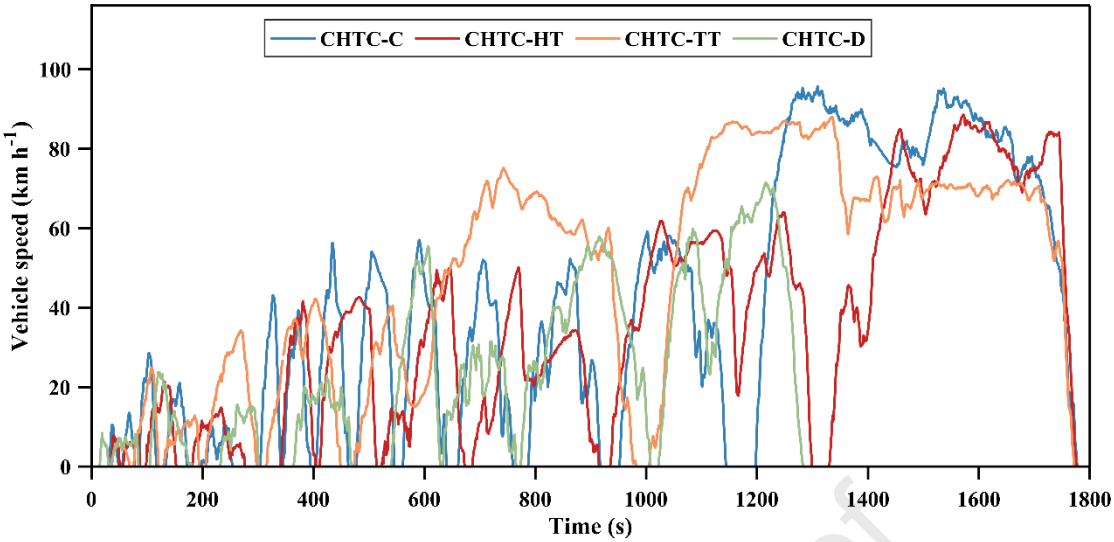


Fig.2. Speed profiles of the four CHTC cycles used for the tested heavy-duty vehicles.

The speed profiles of the four CHTC cycles used in this study are shown in Fig.2. The four cycles represent different operating patterns for different heavy-duty vehicle categories. CHTC-D and CHTC-TT contain two main operating stages, corresponding to low-speed and high-speed operation, whereas CHTC-HT and CHTC-C include more distinct urban, suburban and highway segments. These differences in speed trajectory, acceleration frequency and high-speed duration provide the basis for the vehicle-specific stage division used in the subsequent emission analysis.

2.3 Emission factor calculation

Second by second emission rates were integrated over the whole cycle or over each driving stage to obtain total emissions of CO, NOx and PN. For CO and NOx, the integrated emissions were converted to mass-based units. For PN, the particle number was retained. The distance-based emission factor of pollutant (i) was calculated as:

$$EF_{km,i} = \frac{M_i}{D}$$

where $EF_{km,i}$ is the distance-based emission factor of pollutant (i), M_i is the total emission of pollutant (i) during the corresponding cycle or stage, and (D) is the travelled distance. The units of CO and NOx were mg/km, while PN was expressed as #/km.

The work-based emission factor was calculated as:

$$EF_{kwh,i} = \frac{M_i}{W}$$

where $EF_{kwh,i}$ is the work-based emission factor of pollutant (i), and (W) is the engine work during the corresponding cycle or stage. The units of CO and NOx were mg/kWh, while PN was expressed as #/kWh.

To compare the emission intensity among vehicles with different masses, the vehicle mass normalized distance-based emission factor was further calculated as:

$$EF_{km,t,i} = \frac{EF_{km,i}}{m}$$

where $EF_{km,t,i}$ is the vehicle mass normalized distance-based emission factor of pollutant (i), and (m) is the vehicle mass.

To quantify the emission response induced by air-conditioning operation under each temperature condition, and to provide a basis for evaluating the interaction between high temperature and air-conditioning, the relative change rate was calculated as:

$$R_{air,i} = \left(\frac{EF_{on,i} - EF_{off,i}}{EF_{off,i}}\right) \times 100\%$$

where $R_{air,i}$ is the relative change rate of pollutant (i) caused by air-conditioning operation, $EF_{on,i}$ is the emission factor under air-conditioning on conditions, and $EF_{off,i}$ is the emission factor under air-conditioning off conditions.

To quantify the change in the signed emission response to air-conditioning operation between the two temperatures, the interaction index was defined as:

$$I_i = R_{air,42℃,i} - R_{air,32℃,i}$$

where I_i is the interaction index of pollutant(i), $R_{air,42℃,i}$ is the air-conditioning induced change rate at 42℃, and $R_{air,32℃,i}$ is the air-conditioning induced change rate at 32℃. A positive value of I_i indicates a more positive signed air-conditioning response at 42 ℃ than at 32 ℃, while a negative value indicates a more negative response. The index is expressed in percentage points (pp), because it is the difference between two percentage changes; the AC-induced relative changes themselves are expressed in percent.

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2.4 Machine learning based sensitivity analysis

To further identify variables associated with pollutant responses under different temperature and air-conditioning scenarios, an interpretable machine learning analysis was performed. The machine learning analysis was not intended to develop a generalized fleet emission prediction model. Instead, it was used as a scenario specific sensitivity analysis to identify the variables most closely associated with the second-by-second variations of selected pollutants.

Based on the interaction effect results, three representative vehicle pollutant pairs were selected: Vehicle B's CO, Vehicle E's NOx and Vehicle F's PN. Vehicle B's CO represented a negative high temperature modulation case, in which high temperature weakened the air-conditioning related CO response. Vehicle E–NOx and Vehicle F–PN represented the strongest positive modulation cases for NOx and PN, respectively. For each selected case, four separate models were developed under the four temperature and air-conditioning scenarios.

CatBoost regression models were used to establish nonlinear relationships between input variables and the target pollutant emission rate. For the CO model, torque, NOx, PN, vehicle speed and acceleration were used as input variables. For the NOx model, torque, CO, PN, vehicle speed and acceleration were used. For the PN model, torque, CO, NOx, vehicle speed and acceleration were used. In this framework, speed and acceleration represented vehicle operating state, torque represented engine load, and the other pollutants were treated as co emitted indicators rather than causal explanatory variables.

Model performance was evaluated using the coefficient of determination. SHapley Additive exPlanations were then used to interpret the models. The mean absolute SHAP value of each feature was calculated and normalized to obtain its relative contribution percentage. This method enabled the contribution structure of each target pollutant to be compared among the four scenarios, thereby identifying whether high temperature and air-conditioning operation changed the dominant variables associated with pollutant variations.

3. Results and discussion

3.1 Baseline emission characteristics at the reference temperature with air-conditioning on

To systematically characterize the emission behavior of different heavy duty vehicles, the condition at the reference temperature of 32 ℃ with air conditioning on was selected as the baseline scenario. This condition represents a typical warm ambient operating case and provides a consistent basis for comparing emission characteristics among vehicle categories. Under this baseline condition, the CO, NOx and PN emissions of the six tested heavy-duty vehicles were analyzed.

3.1.1 Emission Characteristics over the Entire Cycle

The whole cycle emission characteristics under the corresponding CHTC cycles were first examined. Fig.3. compares the whole-cycle distance- and work-based emission factors of CO, NOx and PN. Vehicles A-F represented progressively increasing vehicle masses and different vocational categories. A clear pollutant-specific emission pattern was observed among the tested vehicles.

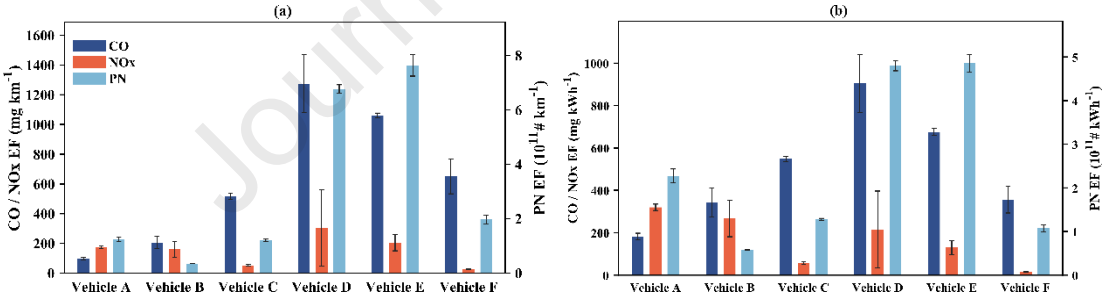


Fig.3. Whole-cycle CO, NOx, and PN emission factors for Vehicles A–F at the reference temperature of 32 ℃ with air conditioning on: (a) distance-based emission factors; (b) work-based emission factors. Bars represent the arithmetic means of two repeated tests, and error bars indicate ±1 sample standard deviation (SD; n = 2).

For CO, Vehicles D and E showed substantially higher distance-based emissions than the other vehicles, and this advantage remained evident after normalization by engine work. Vehicle D is a dump truck and Vehicle E is a freight truck with relatively large vehicle mass, both of which are typically associated with higher traction demand and stronger transient operation during CHTC testing. The elevated CO emissions of

these two vehicles therefore indicate a greater contribution from incomplete combustion or limited oxidation efficiency over the cycle. This interpretation is consistent with previous heavy-duty vehicle studies[16] showing that CO emissions are sensitive to speed range, load fluctuation and aftertreatment thermal conditions, particularly when the diesel oxidation catalyst does not remain in an optimal temperature window.

The NOx pattern differed from that of CO. Vehicles with high CO emissions did not always show the highest NOx emissions, indicating that NOx was controlled by a different balance between engine-out formation and SCR conversion. Tailpipe NOx from modern diesel heavy-duty vehicles is strongly affected by exhaust temperature, urea dosing and SCR activity[17]. Low-speed or transient operation can lower SCR temperature and reduce NOx conversion, whereas sustained higher-load operation may improve the thermal state of the SCR system, even though engine-out NOx formation may increase. In the present results, the relatively high NOx level of Vehicle D suggests that its dump-truck-type operation may not have provided sufficiently stable SCR conditions throughout the cycle. By contrast, Vehicle F, a tractor-trailer with the largest vehicle mass, did not show the highest NOx emission factor, especially on a work basis. This suggests that its larger engine work and more stable high-load operation may have partly offset its distance-based emission intensity through improved SCR thermal conditions.

PN emissions were mainly concentrated in Vehicles D and E, especially in the distance-based results. Since modern diesel heavy-duty vehicles generally rely on DPFs for particle control, high PN emissions are more likely to be related to transient particle formation, DPF filtration efficiency, soot loading or regeneration behavior than to vehicle mass alone. The relatively low PN levels of Vehicles A, B, C and F suggest more stable particle control during the tested cycles, whereas the high PN emissions from D and E may indicate less favorable DPF operating conditions or stronger engine-out particle formation under their corresponding duty cycles. Previous studies[19] have shown that DPF regeneration and aftertreatment operating state can cause substantial variations in PN emissions, even for vehicles equipped with advanced aftertreatment

systems.

The comparison between distance-based emission factors and work-based emission factors provides a more informative interpretation of vehicle differences. The results show that Vehicles D and E had high distance-based emission factors values for CO and PN, indicating their larger contribution to distance-based road emissions. However, after normalization by engine work, the difference among vehicles was partly reshaped. For example, Vehicle F, although having the largest mass, did not show the highest work-based emission factors, suggesting that its larger engine work and relatively stable operating condition diluted its unit-work emission intensity. Conversely, Vehicles D and E still maintained relatively high work-based emission factors for CO and PN, implying that their high emissions were not only caused by greater transport work or vehicle size, but were also associated with less favorable combustion or aftertreatment performance during the tested cycle. Therefore, distance-based emission factors and work-based emission factors should be interpreted as complementary indicators: the former quantifies the real-world road emission burden, whereas the latter provides a more direct assessment of engine- and aftertreatment-related emission behavior.

3.1.2 Emission Characteristics under Different Driving Stages

Following the whole-cycle comparison, the emissions were further resolved by driving stages to identify the road-operation segments contributing to elevated emission intensities. As shown in Fig.4, the stage-resolved analysis was conducted using distance-based emission factors, because distance-based emission factors can more directly characterize the emission burden per unit distance within specific road segments. This metric is particularly suitable for comparing urban, suburban, low-speed and high-speed stages, where differences in average speed, acceleration frequency and travelled distance can markedly reshape the apparent emission intensity.

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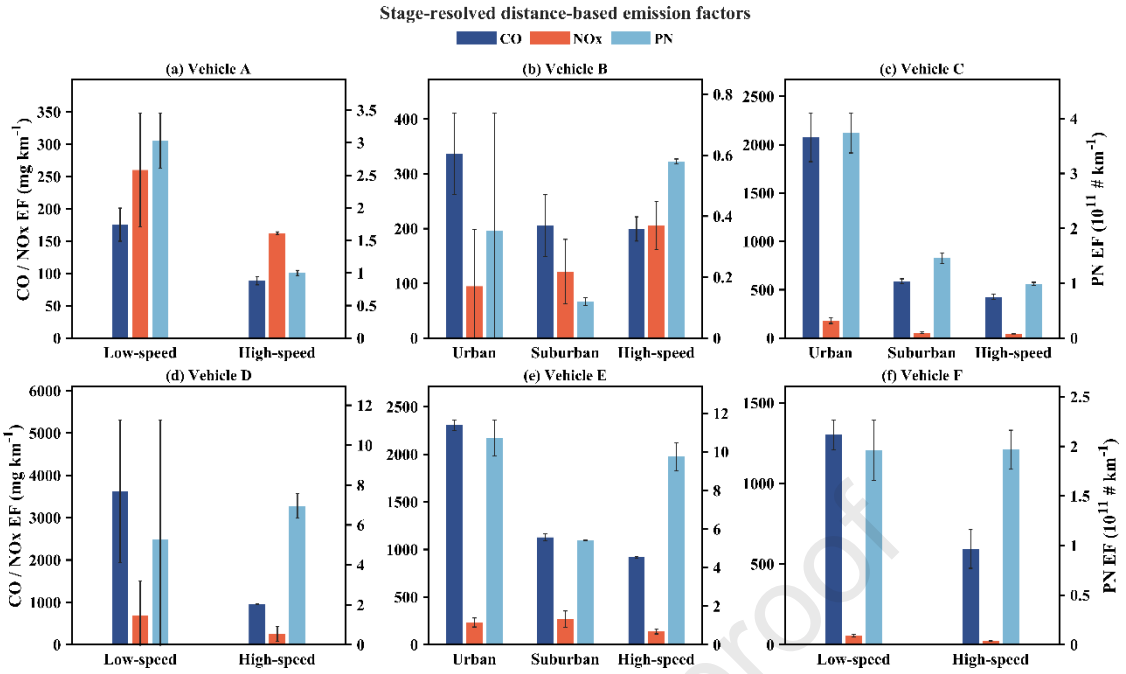


Fig.4. Stage-resolved distance-based emission factors of CO, NOx, and PN for Vehicles A–F at the reference temperature of 32 ℃ with air conditioning on. Bars represent the arithmetic means of two repeated tests, and error bars indicate ±1 sample standard deviation (SD; n = 2).

For CO, the low-speed or urban phases generally exhibited higher emission factors than the high-speed phases. This pattern was particularly evident for Vehicles A, C, D, E and F. Vehicle D showed the most pronounced stage contrast, with the low-speed CO emission factor several times higher than that in the high-speed phase. Similar trends were also observed for Vehicles C and E, whose urban CO emissions were clearly higher than those in suburban and high-speed phases. These results indicate that CO emissions were mainly concentrated in operating phases characterized by low average speed, frequent acceleration and relatively unstable engine loads. Under such conditions, incomplete combustion can be enhanced, while the diesel oxidation catalyst may not maintain a sufficiently stable thermal state for effective CO oxidation. The large error bar for the low-speed CO emission of Vehicle D further suggests that transient operation in this phase introduced considerable repeat-to-repeat variability.

The stage dependence of NOx was more heterogeneous than that of CO. For Vehicles A and D, NOx emissions were higher in the low-speed phase, indicating that

low-speed transient operation may have limited the effectiveness of SCR-based NOx reduction. This is consistent with previous findings that low-load and low-speed operation remains challenging for SCR systems because exhaust temperature and urea dosing conditions are often less favorable. In contrast, Vehicle B showed an increasing NOx tendency from urban/suburban operation to the high-speed phase, and Vehicle E exhibited relatively high NOx in the suburban phase. These results suggest that stage-level NOx emissions reflect a competition between engine-out NOx formation and SCR conversion efficiency. Higher-load stages may increase combustion temperature and engine-out NOx, but they can also improve SCR thermal activity. Consequently, the final tailpipe NOx level depends on which effect dominates during a given stage.

PN emissions showed a distinct stage pattern compared with CO and NOx. For Vehicles C and E, the urban phase contributed the highest PN emission factor, implying that transient acceleration and low-speed operation promoted particle formation or reduced the stability of particle filtration. For Vehicle B, PN was highest in the high-speed phase, while Vehicle D also exhibited higher PN in the high-speed phase than in the low-speed phase. This behavior differs from the CO trend and suggests that PN cannot be interpreted only by combustion completeness. Instead, PN emissions are highly sensitive to DPF filtration status, soot loading, exhaust temperature and possible regeneration-related events. High-speed or high-load operation may increase exhaust temperature and trigger passive or active regeneration, during which particle number emissions can increase markedly even for DPF-equipped diesel vehicles.

Overall, the stage-resolved results indicate that the dominant emission stages differed among pollutants and vehicles under the tested CHTC conditions. CO tended to be higher in low-speed or urban stages for most vehicles, suggesting that incomplete combustion and less stable DOC oxidation may deserve more attention in these operating segments. NOx did not show a uniform stage pattern: higher values appeared in the low-speed stage for some vehicles, while others showed higher NOx in suburban or high-speed operation, implying that the balance between engine-out NOx formation and SCR conversion may vary with vehicle type and stage condition. PN also exhibited vehicle-specific stage peaks, with some vehicles showing higher PN in urban operation

and others in high-speed operation, which may be related to transient particle formation, DPF filtration status, or possible regeneration-related behavior.

3.1.3 Vehicle Mass-Normalized Emission Characteristics

Based on the whole-cycle and stage-resolved emission characteristics discussed above, the distance-based emission factors were further normalized by vehicle mass to examine whether the apparent emission differences remained after accounting for vehicle size. The results are shown in Fig.5.

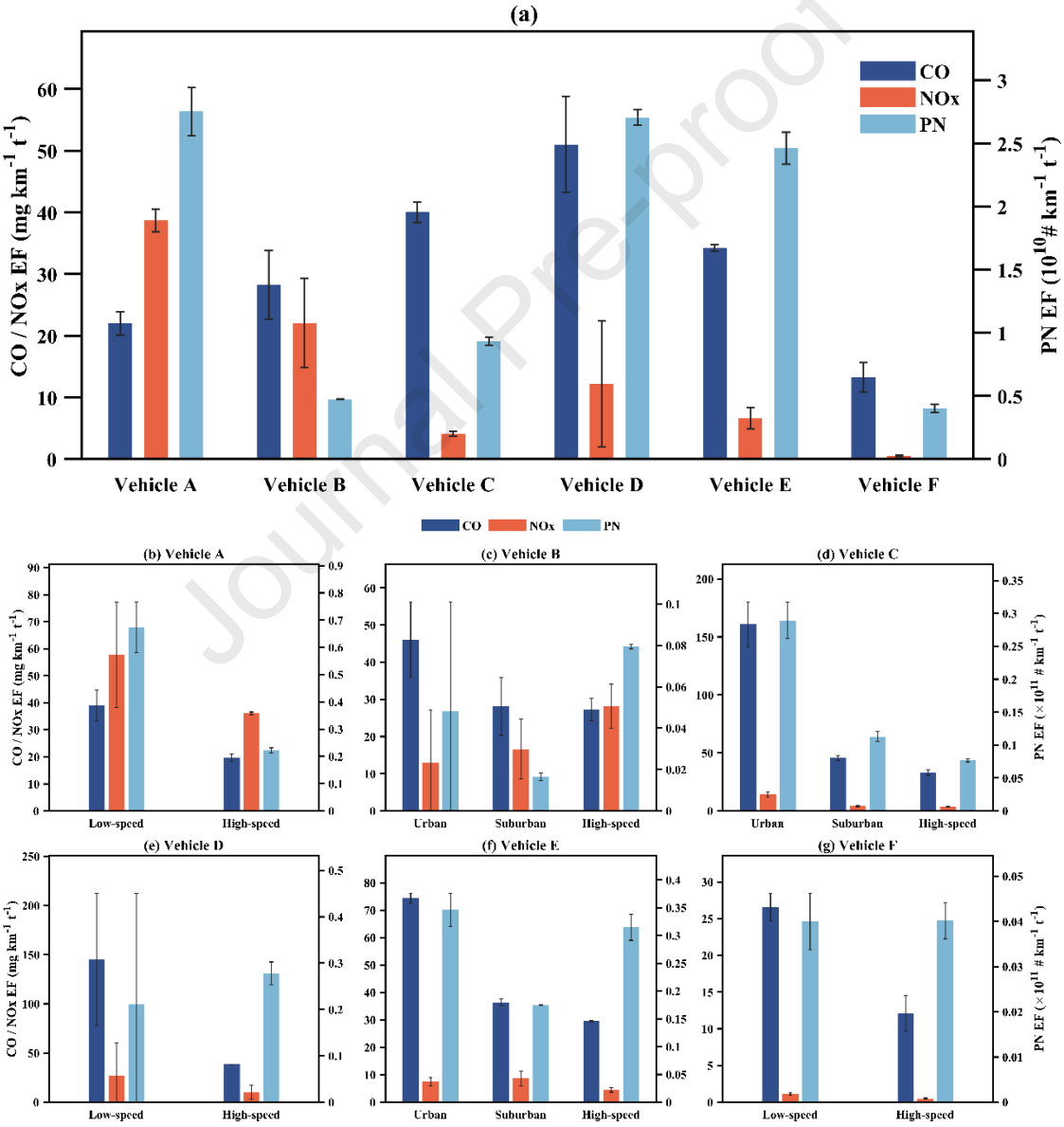


Fig.5. Vehicle-mass-normalized distance-based emission factors of CO, NOx, and PN at the reference temperature of 32 ℃ with air conditioning on: (a) whole-cycle results;

(b-g) stage-resolved results for Vehicles A-F, respectively. Bars represent the arithmetic means of two repeated tests, and error bars indicate ±1 sample standard deviation (SD; n = 2).

The vehicle-mass-normalized results changed the interpretation obtained from the absolute distance-based emissions. Vehicle F, the heaviest tractor-trailer, showed the lowest normalized emission levels for most pollutants, especially for CO and NOx. This indicates that its large vehicle mass and relatively stable operating pattern diluted the emission burden when emissions were expressed per unit vehicle mass. In contrast, Vehicle A, the lightest vehicle in this test group, showed the highest normalized NOx and PN emissions, although its absolute emissions were not the highest in the whole-cycle comparison. Vehicle C also exhibited relatively high normalized CO emissions. These results suggest that smaller or medium-weight heavy-duty vehicles may have higher emission intensity per unit vehicle mass under certain operating conditions.

The pollutant-specific results provide different implications. For CO, Vehicle D remained the highest after mass normalization, while Vehicle C also became prominent. This indicates that the high normalized CO emissions were not simply a result of larger vehicle size, but were more closely related to the efficiency of converting fuel energy into useful transport work under the tested cycle. For NOx, Vehicle A had the highest normalized value, followed by Vehicle B and Vehicle D, suggesting that light-duty operation within the heavy-duty category may lead to a higher NOx burden per unit vehicle mass when aftertreatment performance is not sufficiently stable. For PN, Vehicle A also showed the highest normalized value, whereas Vehicle F remained low, implying that particle-control stability may be more favorable for the tractor-trailer under the present test condition.

The stage-resolved normalized results further show that the mass-specific emission intensity was concentrated in different stages for different vehicles. Vehicle A showed high normalized NOx and PN in the low-speed stage, Vehicle C showed high normalized CO and PN in the urban stage, and Vehicle D presented high normalized CO in the low-speed stage with large test-to-test variability. In comparison, Vehicle F

maintained relatively low normalized values in both low-speed and high-speed stages. This pattern suggests that the potential benefit of larger-capacity vehicles may be more apparent under stable operating conditions, whereas low-speed or urban operation may weaken such benefits if combustion efficiency or aftertreatment operation becomes less favorable.

From the perspective of transport-structure optimization, the mass-normalized results suggest that larger-capacity vehicles may have advantages in reducing emissions per unit vehicle mass when they are operated under suitable duty cycles. Vehicle F, for example, showed consistently low mass-normalized emissions, indicating that tractor-trailer operation with relatively stable load and speed conditions may provide better emission efficiency than smaller vehicles for long-distance or high-capacity freight transport. This implies that, for freight activities with sufficient cargo demand, replacing multiple small- or medium-capacity trips with fewer high-capacity trips could potentially reduce the emission intensity per unit transport capacity. However, this advantage depends on route characteristics, load factor and aftertreatment stability. The relatively high mass-normalized CO or PN emissions of Vehicles D and E indicate that larger vehicle size alone does not guarantee lower normalized emissions, particularly under low-speed, urban or highly transient operation. Therefore, transport optimization should not simply favor larger vehicles, but should match vehicle capacity with operating scenarios: high-capacity vehicles may be more suitable for stable freight corridors, whereas vehicles frequently operating in low-speed or urban conditions require greater attention to combustion efficiency and aftertreatment control.

3.2 Effects of air-conditioning operation on emissions

Vehicle cabin thermal management studies[6] have shown that air-conditioning systems impose a considerable auxiliary power demand on vehicles and can reduce fuel economy, especially when cabin heat loads increase under warm ambient conditions. Therefore, this section further evaluates how air-conditioning operation affected the CO, NOx and PN emissions of the tested heavy-duty vehicles at the reference temperature conditions, from whole-cycle, stage-resolved and transient perspectives.

3.2.1 Whole-cycle effects

The whole cycle influence of air-conditioning operation at the reference temperature is presented in Fig.6 and Fig.7. Fig.6 shows the paired changes in distance-based emission factors before and after air-conditioning operation, and Fig.7 further compares the corresponding relative changes in distance-based and work-based emission factors.

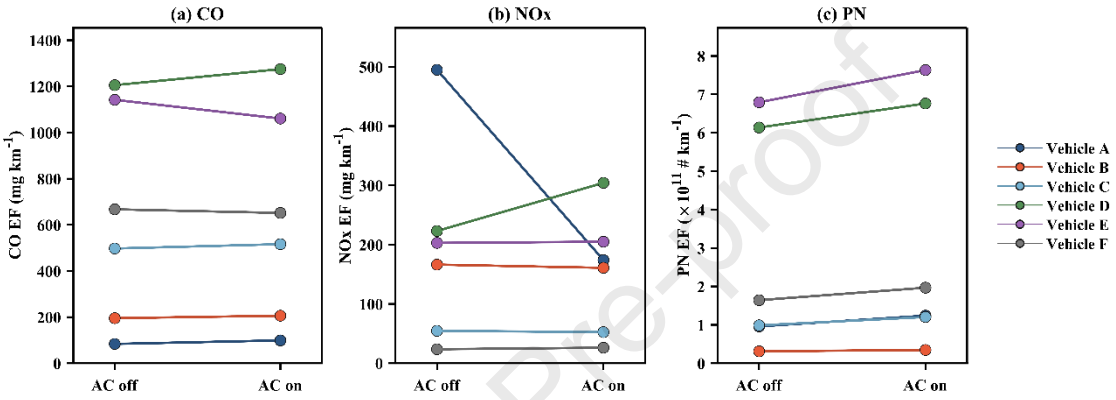


Fig.6. Air-conditioning-induced changes in distance-based CO, NOx and PN emission factors at the reference temperature conditions.

For CO, the distance-based emission factor increased slightly for Vehicles A to D after air-conditioning was turned on, but decreased for Vehicles E and F. The absolute increase was small for Vehicles A and B, while Vehicle D maintained the highest CO level under both air-conditioning off and on conditions. The decrease observed for Vehicles E and F indicates that the additional load from air-conditioning did not necessarily lead to higher tailpipe CO emissions. A possible explanation is that the additional engine load raised exhaust temperature and improved the oxidation activity of the diesel oxidation catalyst, partly offsetting the increase in engine out CO formation. Therefore, the CO response can be interpreted as the combined result of combustion related CO formation and catalyst oxidation efficiency, rather than as a direct consequence of the additional auxiliary load.

The NOx response showed a different pattern. Vehicle A exhibited a large reduction in distance-based NOx after air-conditioning operation, whereas Vehicle D

showed a clear increase. The other vehicles changed only slightly. This opposite response is consistent with the dual influence of air-conditioning on NOx emissions. Increased engine load can raise combustion temperature and promote engine out NOx formation; at the same time, higher exhaust temperature may improve the activity of the selective catalytic reduction system and increase NOx conversion. For Vehicle A, the reduction in tailpipe NOx suggests that the improvement in aftertreatment conditions may have been stronger than the increase in engine out NOx formation. For Vehicle D, the increase in NOx suggests that the additional formation of NOx or the limited compensation by the selective catalytic reduction system may have dominated during the cycle.

PN increased for all vehicles after air-conditioning operation. This response was more consistent than that of CO and NOx. Vehicles D and E had substantially higher absolute PN levels than the other vehicles, and their PN emissions increased further after air-conditioning was turned on. This suggests that particle number emissions were more sensitive to the extra transient load introduced by air-conditioning.

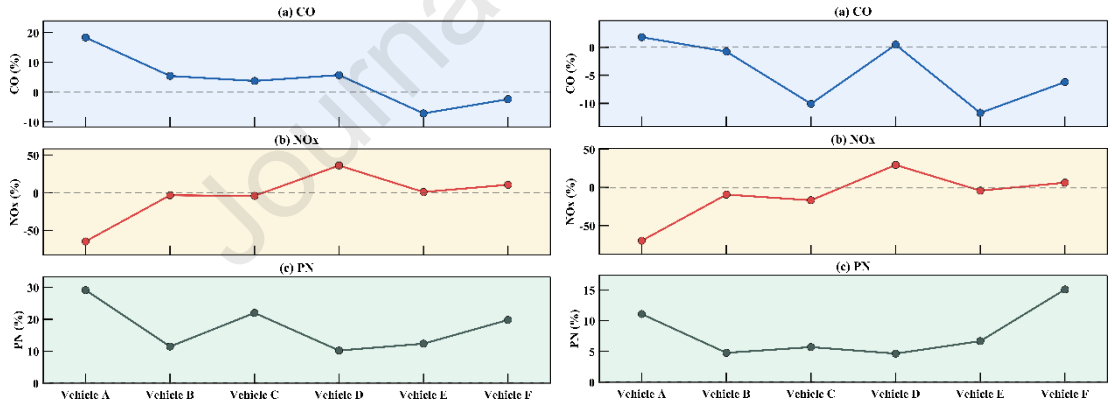


Fig.7. Relative changes in emission factors caused by air-conditioning operation at the reference temperature conditions: distance-based emission factors on the left and work-based emission factors on the right.

Fig.7 further compares the relative changes in distance-based and work-based emission factors caused by air-conditioning operation. Since Vehicles A to F correspond to increasing vehicle mass, the figure also provides a basis for examining whether the air-conditioning effect varied with vehicle size.

For CO, the distance-based change rate generally decreased from the lighter vehicles to the heavier vehicles. Vehicles A to D showed positive changes, whereas Vehicles E and F showed negative changes. After normalization by engine work, the CO response became smaller for most vehicles and turned negative for Vehicles C, E and F. This indicates that the increase in distance-based CO for some vehicles was partly associated with the additional work demand introduced by air-conditioning, rather than a proportional increase in emission intensity per unit engine output. The negative work-based changes for the heavier vehicles suggest that air-conditioning did not aggravate their intrinsic CO emission intensity under the tested condition.

For NOx, the response did not follow a monotonic trend with vehicle mass. Vehicle A showed the largest reduction in both distance-based and work-based metrics, whereas Vehicle D showed the largest increase. Vehicles B, C and E changed only slightly, and Vehicle F showed a moderate positive response. This result suggests that the effect of air-conditioning on NOx was not primarily controlled by vehicle mass.

For PN, all vehicles showed positive relative changes after air-conditioning operation. However, the magnitude of the increase was lower in the work-based metric than in the distance-based metric for most vehicles. This difference indicates that part of the PN increase per unit distance was related to the additional engine work demand. Nevertheless, the work based PN changes remained positive for all vehicles, suggesting that air-conditioning operation still increased PN emission intensity per unit engine output. Unlike CO, the PN response did not decrease clearly with increasing vehicle mass; both the lightest vehicle and the heaviest vehicle showed relatively evident positive changes.

Overall, air-conditioning operation produced different response patterns among the three pollutants at the reference temperature. CO showed limited changes, with slight increases for Vehicles A–D and decreases for Vehicles E and F, and the work based results further weakened this response. NOx presented the most pronounced contrast among vehicles, with a substantial decrease for Vehicle A but an increase for Vehicle D, while the other vehicles remained relatively stable. PN was the only pollutant that increased for all vehicles, and its positive response was retained after

normalization by engine work, although the magnitude was generally reduced. In terms of vehicle mass, the air-conditioning effect on CO tended to weaken as vehicle mass increased, whereas NOx did not follow a clear mass related trend. PN increased across the entire vehicle sequence from A to F, indicating that particle number emissions were more consistently affected by air-conditioning operation than CO and NOx in the whole cycle results.

3.2.2 Stage resolved effects

To further identify the operating stages responsible for the whole cycle changes discussed above, the relative changes in stage resolved emission factors after air-conditioning operation were calculated, as shown in Fig.8. The upper row presents the distance-based results, and the lower row presents the work-based results. Positive values indicate that air-conditioning increased the corresponding emission factor, while negative values indicate a reduction. Blank cells represent stages that were not included in the corresponding vehicle cycle.

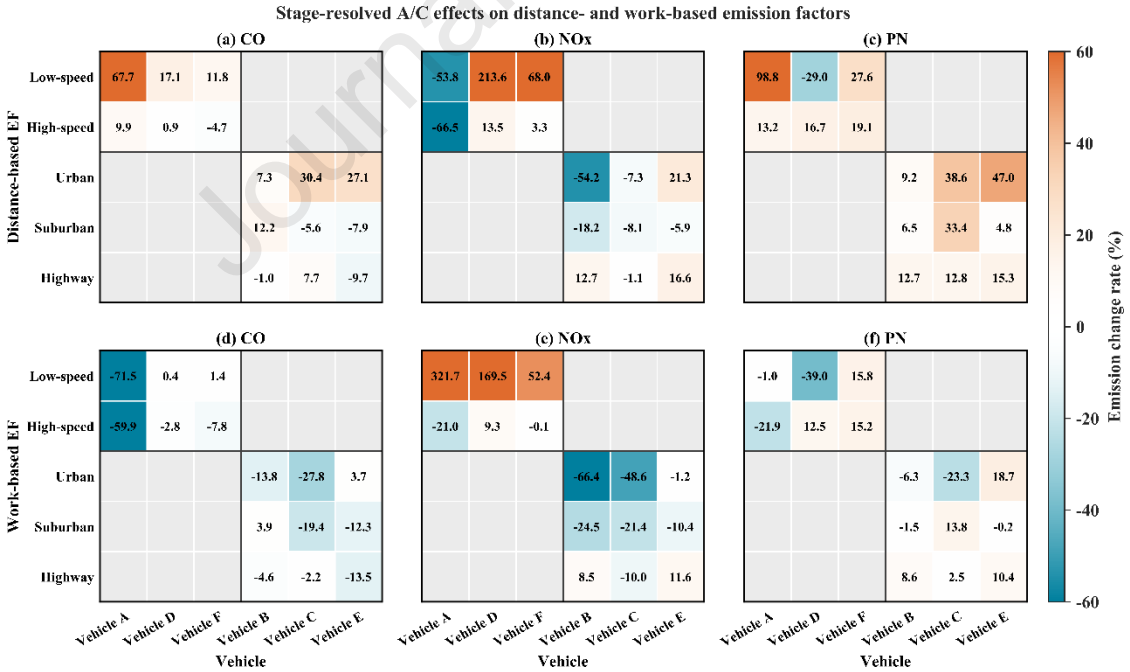


Fig.8. Stage-resolved relative changes in CO, NOx, and PN emission factors induced by air-conditioning operation at the reference temperature of 32 ℃. Panels (a–c) show distance-based results for CO, NOx, and PN, respectively; panels (d–f) show the corresponding work-based results. Changes are calculated relative to the AC-off

565 emission factor at 32 ℃ and expressed as percentages; positive and negative values
566 indicate increases and decreases, respectively. Blank gray cells indicate stages not
567 included in the corresponding vehicle-specific CHTC cycle.
568
569 For CO, the distance-based results show that the increase caused by air-
570 conditioning was mainly concentrated in low speed or urban operation. Vehicle A
571 showed the largest increase in the low-speed stage, reaching 67.7%, while Vehicles C
572 and E showed increases of 30.4% and 27.1% in the urban stage, respectively. In contrast,
573 the high speed or highway stages generally showed smaller changes, and negative
574 values appeared for Vehicle F in the high-speed stage and for Vehicles B and E in the
575 highway stage. This indicates that the whole cycle CO increase observed for some
576 vehicles was mainly associated with low speed or urban segments rather than uniformly
577 distributed across the entire cycle.
578 After normalization by engine work, the CO response was markedly weakened.
579 Vehicle A changed from strong positive distance-based increases to negative work-
580 based changes in both the low speed and high-speed stages. Vehicles C and E also
581 showed negative work-based changes in several stages. This contrast suggests that a
582 considerable part of the distance-based CO increase was related to the additional work
583 demand introduced by air-conditioning, whereas the CO emission intensity per unit
584 engine output did not increase proportionally. For heavier vehicles, especially Vehicles
585 E and F, the work-based CO changes were mostly small or negative, which is consistent
586 with the whole cycle result that air-conditioning did not aggravate their CO emission
587 intensity.
588 For NOx, the stage pattern was more concentrated and more variable. In the
589 distance-based results, Vehicle D showed the strongest increase in the low-speed stage,
590 with a change rate of 213.6%, followed by Vehicle F in the low-speed stage at 68.0%.
591 In contrast, Vehicle A showed reductions in both low speed and high-speed stages. For
592 Vehicles B, C and E, the urban and suburban stages were mostly negative or weakly
593 positive, while the highway stage showed small positive changes for Vehicles B and E.
594 These results indicate that the overall NOx response was strongly influenced by a

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limited number of stage specific changes, especially the low-speed response of Vehicle D.

The work-based NOx results further highlight the importance of the low-speed stage. Vehicles A, D and F all showed positive work-based NOx changes in the low-speed stage, with particularly large increases for Vehicles A and D. However, many urban and suburban stages for Vehicles B, C and E remained negative after work normalization. This means that the NOx response after air-conditioning operation cannot be explained only by the increase in engine work. Instead, the direction and magnitude of NOx changes varied substantially among stages, and the low-speed segments of the two stage cycles were the most sensitive in the work-based results.

For PN, the distance-based response was generally positive. Vehicle A showed a large increase of 98.8% in the low-speed stage, while Vehicles C and E showed clear increases in the urban stage, reaching 38.6% and 47.0%, respectively. Most high speed and highway stages also showed positive changes, although the magnitudes were smaller. The main exception was Vehicle D in the low-speed stage, where distance based PN decreased by 29.0%. These results are consistent with the whole cycle observation that PN was the pollutant with the most consistent positive response to air-conditioning operation.

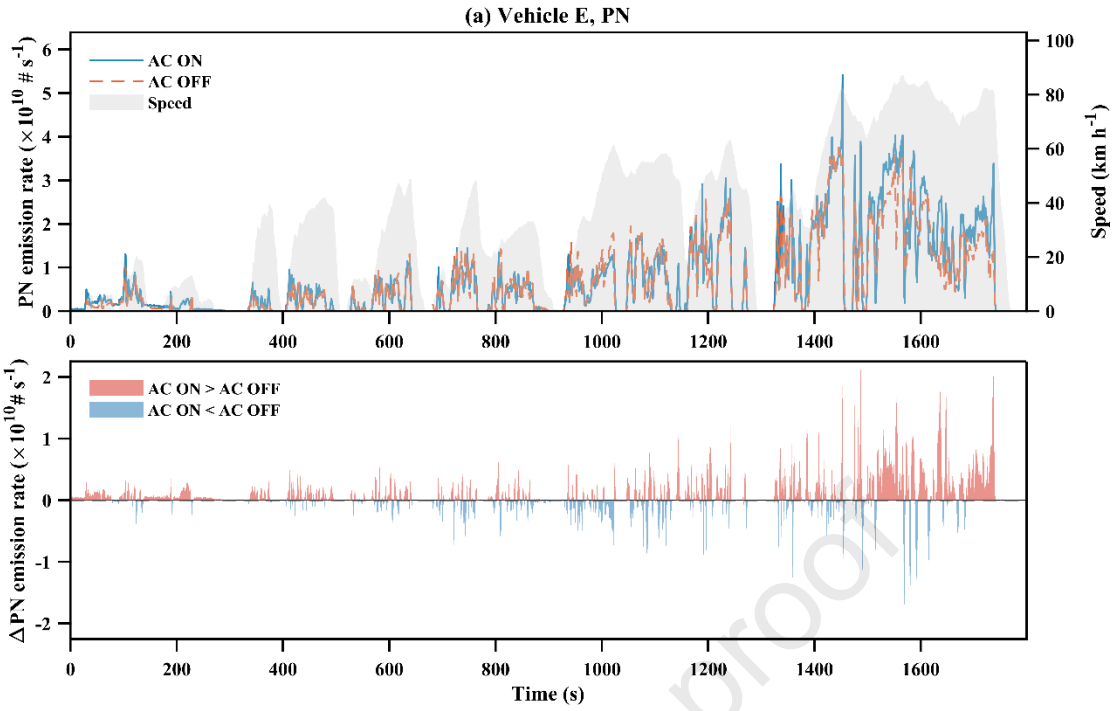
The work based PN results were more mixed than the distance-based results. Some positive responses remained, such as Vehicle F in both low speed and high-speed stages, Vehicle D in the high-speed stage, and Vehicle E in the urban and highway stages. However, several distance-based increases became smaller or turned negative after work normalization, including Vehicle A in both stages and Vehicle C in the urban stage. This suggests that part of the PN increase per unit distance was associated with the additional work required by air-conditioning, while some stages still showed an increase in particle number emission intensity per unit engine output.

Overall, Fig.8 shows that the whole cycle effects of air-conditioning were produced by different stage contributions for different pollutants. For CO, the distance-based increases were mainly observed in low speed or urban stages, but these increases were substantially weakened or reversed after normalization by engine work, indicating

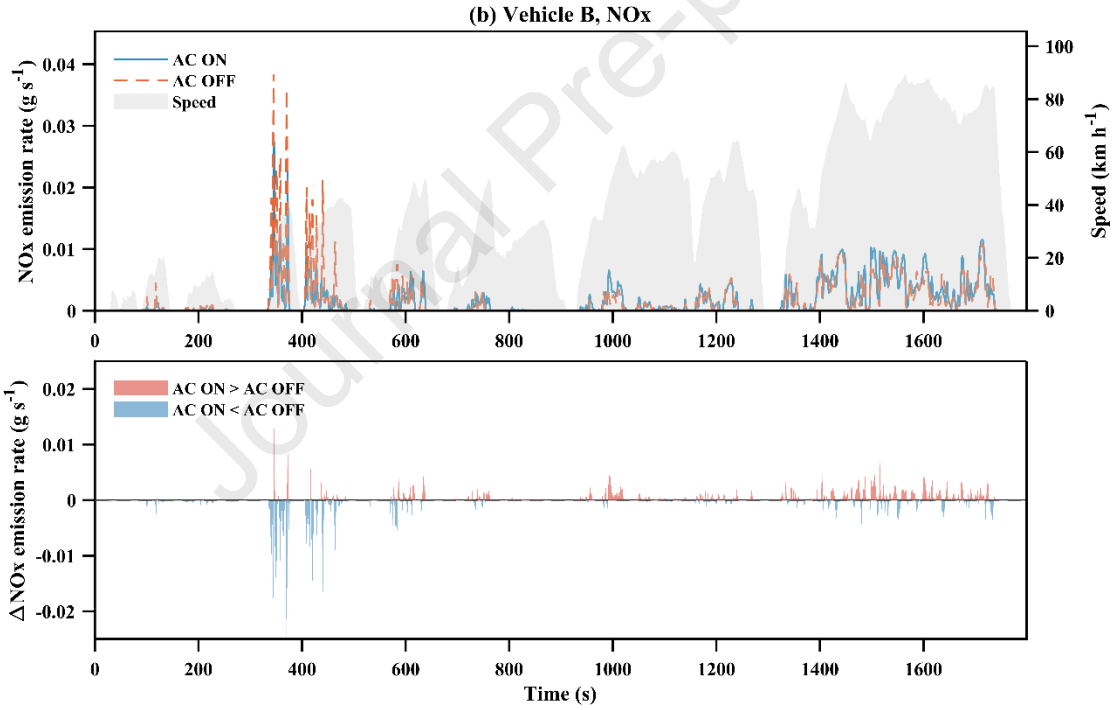
that the CO response was closely related to the additional work demand in these stages. For NOx, the response was more concentrated in specific vehicle stage combinations, with the strongest positive changes occurring in the low-speed stage of Vehicles D and F, whereas several urban and suburban stages showed negative changes. For PN, positive changes appeared in most distance-based stage results, particularly in the low-speed stage of Vehicle A and the urban stages of Vehicles C and E; however, some of these increases became weaker or negative in the work-based results. These patterns indicate that air-conditioning operation affected CO, NOx and PN through different stage dependent responses rather than through a uniform whole cycle shift.

3.2.3 Transient characteristics

To further examine how the whole cycle and stage resolved changes were formed, representative second by second emission profiles were selected for detailed analysis, as shown in Fig.9. The selected cases include Vehicle E for PN, Vehicle B for NOx and Vehicle F for CO. These cases were chosen because they represent three different response patterns observed in the previous results: a pollutant with a generally positive response under air-conditioning operation, a pollutant with alternating positive and negative responses, and a pollutant whose response was dominated by frequent transient fluctuations. The upper panel of each case compares the emission rates under air-conditioning on and off conditions together with the speed profile, while the lower panel presents the instantaneous difference between the two conditions. For each selected vehicle-pollutant pair and AC condition at 32 ℃, the plotted emission-rate profile is the pointwise arithmetic mean of the two repeated tests at corresponding times in the prescribed CHTC cycle. The instantaneous difference was calculated as the mean AC-on emission rate minus the mean AC-off emission rate at each corresponding cycle time. Positive and negative differences therefore indicate higher and lower emission rates with air conditioning on, respectively.



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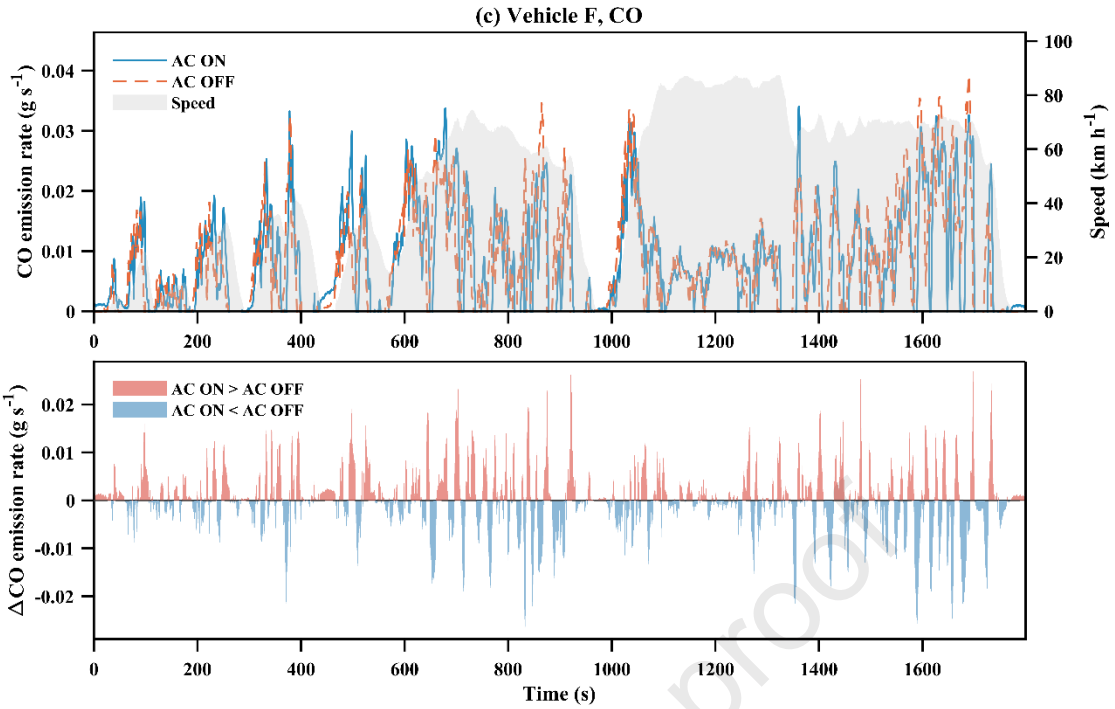


Fig.9. Representative second-by-second mean emission-rate profiles at the reference temperature of 32 ℃ under AC-on and AC-off conditions: (a) PN from Vehicle E; (b) NOx from Vehicle B; (c) CO from Vehicle F. Each emission-rate profile is the pointwise arithmetic mean of two repeated tests. For each case, the upper panel shows the mean emission-rate profiles and the speed profile; the lower panel shows the instantaneous difference calculated as the mean AC-on emission rate minus the mean AC-off emission rate.

For Vehicle E, the PN emission rate under air-conditioning on conditions was frequently higher than that under air-conditioning off conditions, particularly in the later part of the cycle after approximately 1300 s. This period corresponded to relatively high-speed operation with frequent transient fluctuations, during which the difference curve showed repeated positive excursions. Although negative differences also occurred at some moments, the positive peaks were more frequent and larger in magnitude, leading to a net increase in PN. This transient pattern is consistent with the whole cycle and stage resolved results, where PN generally increased after air-conditioning operation. The result suggests that the PN response was not a continuous elevation over the entire cycle, but was formed by repeated short duration increases

673 during specific operating intervals.

674 For Vehicle B, the NOx response was mainly influenced by several transient peaks
675 in the early part of the cycle. Around the first acceleration intensive period, the NOx
676 emission rate under air-conditioning off conditions was higher than that under air-
677 conditioning on conditions, resulting in pronounced negative values in the difference
678 curve. In the later part of the cycle, especially during the high speed section, the two
679 NOx curves became closer and the difference fluctuated around zero. This indicates
680 that the reduction in NOx observed at the whole cycle or stage level was not caused by
681 a uniform decrease throughout the cycle, but was largely associated with a limited
682 number of transient high emission events. The comparison with the speed profile also
683 suggests that the air-conditioning effect on NOx was not determined by speed alone;
684 some rapid changes in operating state produced stronger differences than steady high-
685 speed operation.

686 For Vehicle F, CO showed intensive pulse like behavior under both air-
687 conditioning conditions. The instantaneous difference between air-conditioning on and
688 off conditions alternated frequently between positive and negative values over the cycle.
689 Positive differences appeared during some acceleration related emission peaks, whereas
690 negative differences also occurred in similar operating periods. This explains why the
691 whole cycle CO response was relatively limited despite the presence of many short-
692 term deviations. Compared with PN, the CO response did not show a persistent
693 direction; compared with NOx, it was less dominated by a small number of early peaks.
694 Instead, the net CO effect was determined by the balance between numerous positive
695 and negative transient pulses.

696 Overall, the representative second by second results show that the effects of air-
697 conditioning operation were formed through different transient patterns for different
698 pollutants. PN from Vehicle E was characterized by repeated positive excursions,
699 mainly during the later medium and high-speed portions of the cycle. NOx from Vehicle
700 B was dominated by several early transient peaks where air-conditioning operation
701 reduced the emission rate. CO from Vehicle F showed frequent bidirectional
702 fluctuations, resulting in a moderate net response. These observations indicate that the

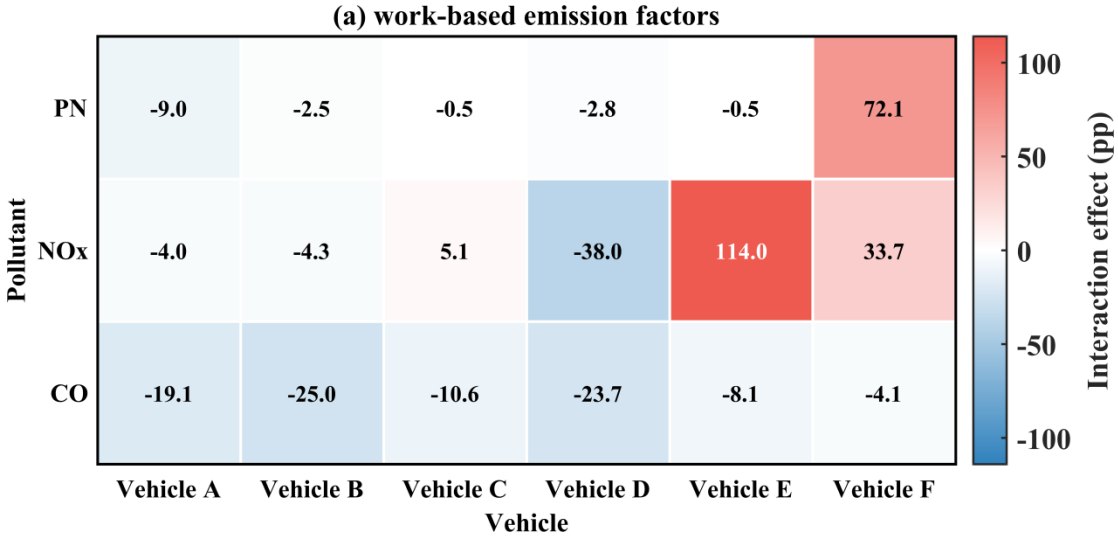
whole cycle changes reported above were not simple shifts of the emission profiles, but the cumulative outcome of short transient responses occurring at different moments of the cycle.

3.3 High temperature modulation of air-conditioning related emission responses and sensitivity identification

This section further examines whether high temperature altered the emission response induced by air-conditioning operation, first quantifies the interaction effect between high temperature and air-conditioning operation, and then uses machine learning based sensitivity analysis to identify variables associated with representative pollutant responses.

3.3.1 Interaction effect between high temperature and air-conditioning operation

To examine how high temperature changed the signed emission response to air-conditioning operation, an interaction index was calculated by subtracting the AC-induced percentage change at 32 ℃ from that at 42 ℃. The index is expressed in percentage points (pp). A positive index indicates a more positive signed AC response at 42 ℃, whereas a negative index indicates a more negative response. Fig.10 presents the work-based results in the upper panel and the distance-based results in the lower panel.



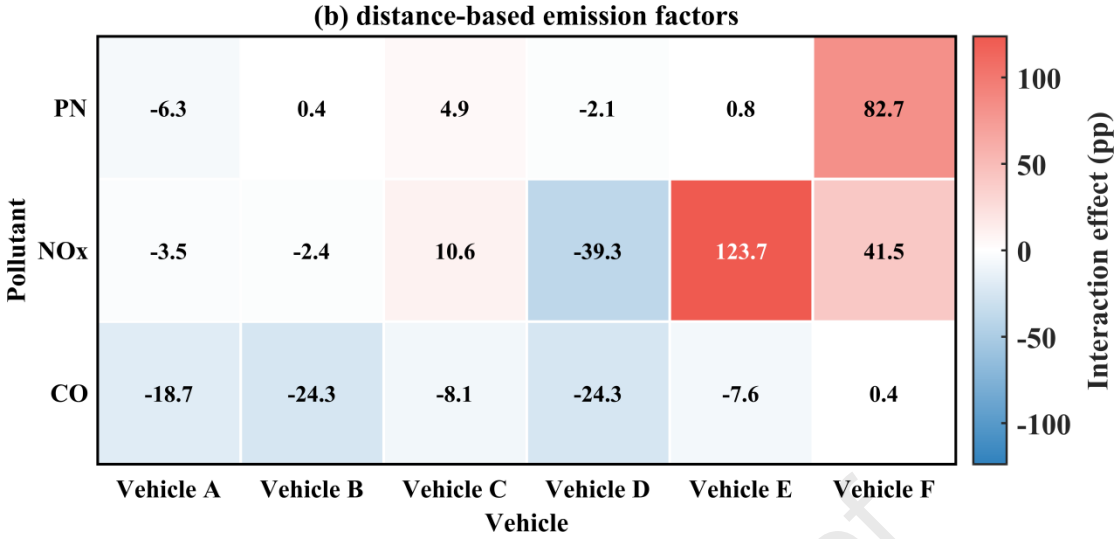


Fig.10. High-temperature modulation of the AC-induced relative changes in whole-cycle CO, NOx, and PN emission factors: (a) work-based results; (b) distance-based results. Each cell shows the AC-induced percentage change at 42 ℃ minus that at 32 ℃, expressed in percentage points (pp).

For CO, the interaction indices were mostly negative in both metrics. In the distance-based results, Vehicles A–E showed values ranging from -24.3 to -7.6 pp, while Vehicle F was close to zero. The work-based indices were negative for all six vehicles, ranging from -25.0 to -4.1 pp. These results indicate that the signed AC-induced CO response was generally lower at 42 ℃ than at 32 ℃, especially for Vehicles B and D. The similar patterns in the distance-based and work-based results indicate consistent negative CO modulation across the two metrics.

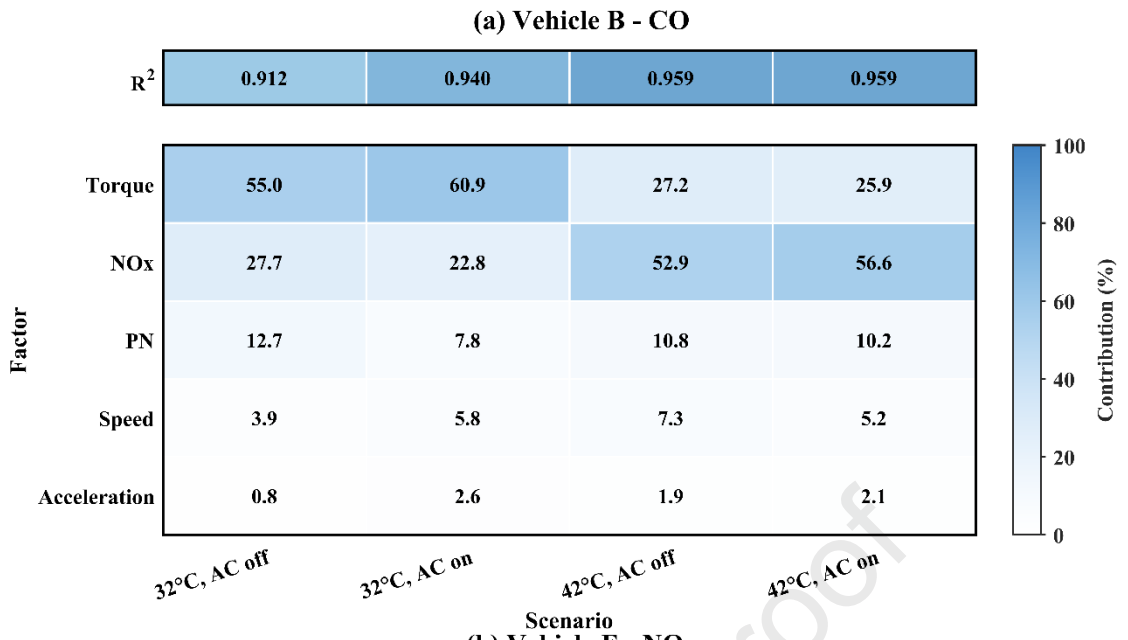
NOx showed a much stronger and more variable interaction pattern than CO. Vehicle E exhibited the largest positive interaction effect, 123.7 pp in the distance-based result and 114.0 pp in the work-based result. Vehicle F also showed positive values in both metrics, while Vehicle C showed only a weak positive response. In contrast, Vehicle D showed a clear negative interaction effect, -39.3 pp in the distance-based result and -38.0 pp in the work-based result. Vehicles A and B remained slightly negative. These results indicate that the high temperature modulation of NOx was highly selective among vehicles, with Vehicle E representing the strongest enhancement and Vehicle D representing a clear weakening effect.

For PN, the most notable positive interaction appeared in Vehicle F, with values of 82.7 pp for the distance-based metric and 72.1 pp for the work-based metric. In comparison, the other vehicles showed relatively small interaction effects, mostly close to zero or slightly negative. This indicates that the high temperature enhancement of the air-conditioning effect on PN was mainly concentrated in Vehicle F rather than being a general response across all vehicles. The agreement between the distance based and work-based results further suggests that the PN enhancement for Vehicle F was not simply due to increased work demand per unit distance, but remained evident after normalization by engine work.

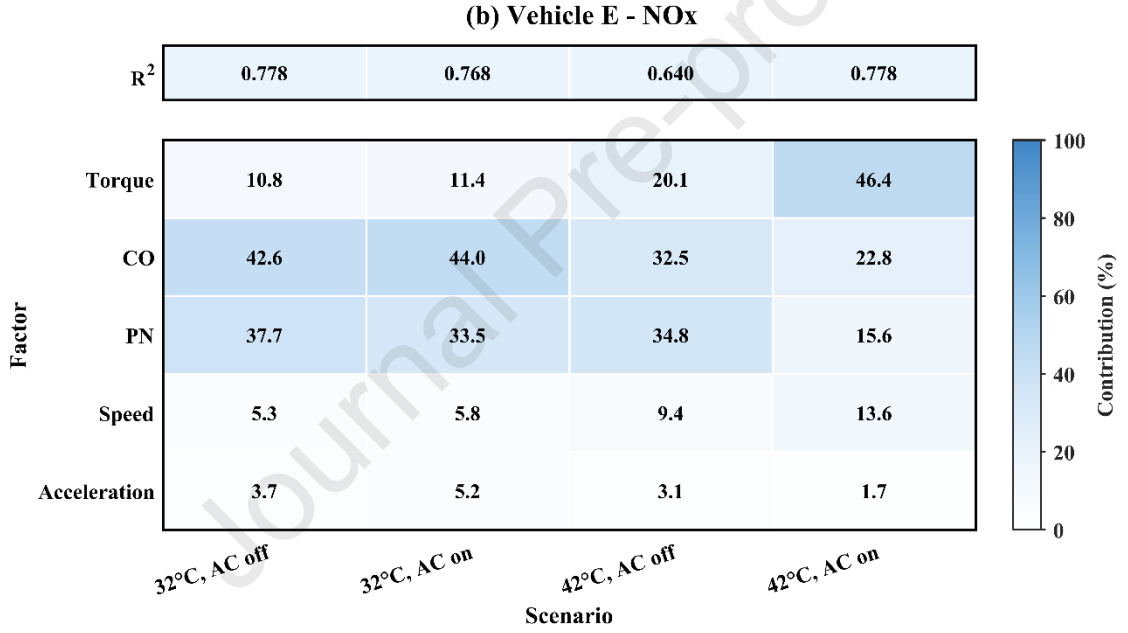
Overall, Fig. 10 shows that high temperature did not uniformly increase the signed emission response to air-conditioning operation across pollutants and vehicles. CO was mainly characterized by negative modulation, NOx showed the strongest vehicle specific contrast, and PN displayed a pronounced positive modulation only for Vehicle F. The similar patterns between work-based and distance-based interaction effects indicate that the observed modulation characteristics were relatively robust to the choice of emission factor metric.

3.3.2 Machine learning based sensitivity identification under four operating scenarios

Based on the interaction effects shown in Fig.10, the upper strip shows the coefficient of determination, R^2, of the CatBoost model under each scenario, and the heatmap values represent normalized mean absolute SHAP contributions of the input variables. Three representative vehicle pollutant pairs were selected for machine learning based sensitivity analysis: Vehicle B for CO, Vehicle E for NOx and Vehicle F for PN. Vehicle B's CO represents a typical negative modulation case, where high temperature weakened the effect of air-conditioning operation on CO emissions. Vehicle E's NOx and Vehicle F's PN represent the strongest positive modulation cases for NOx and PN, respectively. CatBoost models were developed for each temperature and air-conditioning scenario, and SHAP-based feature contributions were used to identify the variables associated with the second-by-second pollutant variations.

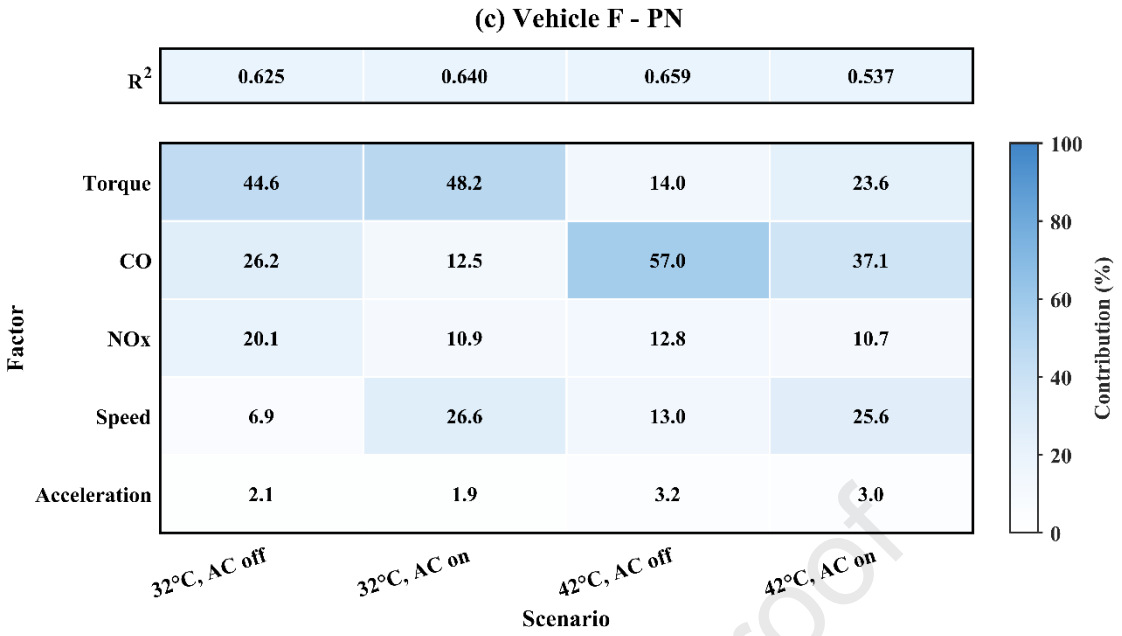


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780 **Fig.11.** CatBoost model performance and SHAP-based feature contributions under four
781 temperature and air-conditioning scenarios: (a) CO from Vehicle B; (b) NOx from
782 Vehicle E; (c) PN from Vehicle F. The upper strips show the coefficient of determination
783 (R^2) for each scenario. Heatmap values are the mean absolute SHAP contributions
784 normalized across the input variables within each model and expressed as percentages.
785 These values represent relative model-based feature contributions and do not establish
786 causal effects.

787

788 For Vehicle B's CO, the model performance was high and stable across the four
789 scenarios, with R^2 values ranging from 0.912 to 0.959. At the reference temperature,
790 torque was the dominant variable, contributing 55.0% under air-conditioning off and
791 60.9% under air-conditioning on conditions. This indicates that CO variation was
792 mainly associated with engine-load-related behavior at the reference temperature, and
793 air-conditioning operation further strengthened the relative contribution of torque.
794 However, under high temperature, the contribution of torque decreased markedly to
795 27.2% and 25.9%, while NOx became the dominant co-emitted indicator, with
796 contributions of 52.9% and 56.6% under air-conditioning off and on conditions,
797 respectively. The contributions of PN, speed and acceleration remained relatively low.
798 This shift suggests that the CO response of Vehicle B under high temperature was no
799 longer primarily associated with the additional load represented by torque, but became

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more strongly synchronized with NOx-related emission behavior.

For Vehicle E's NOx, the model showed moderate explanatory ability, with R^2 values of 0.778, 0.768, 0.640 and 0.778 across the four scenarios. At the reference temperature, CO and PN were the main co-emitted indicators associated with NOx variation, contributing 42.6–44.0% and 33.5–37.7%, respectively, whereas torque contributed only 10.8–11.4%. Under high temperature without air-conditioning operation, CO and PN still contributed substantially, while the torque contribution increased to 20.1%. A more pronounced change occurred under high temperature with air-conditioning operation, where torque became the dominant variable, contributing 46.4%, while the contributions of CO and PN decreased to 22.8% and 15.6%, respectively. Speed also increased to 13.6%. These results indicate that the combined high-temperature and air-conditioning scenario substantially reshaped the NOx sensitivity structure of Vehicle E, shifting it from a co-emission-dominated pattern toward a load-related pattern.

For Vehicle F's PN, the model performance was lower than that of Vehicle B's CO and Vehicle E's NOx, with R^2 values between 0.537 and 0.659. This suggests that PN emissions from Vehicle F were affected by more complex transient processes that were not fully captured by the selected variables. At the reference temperature, torque was the most important variable, contributing 44.6% under air-conditioning off and 48.2% under air-conditioning on conditions. Meanwhile, air-conditioning operation increased the contribution of speed from 6.9% to 26.6%, indicating that the PN response was associated not only with engine load but also with speed-related operating characteristics. Under high temperature without air-conditioning operation, CO became the dominant co-emitted indicator, contributing 57.0%, whereas torque decreased to 14.0%. Under high temperature with air-conditioning operation, CO remained the largest contributor at 37.1%, while torque and speed increased to 23.6% and 25.6%, respectively. These patterns suggest that the PN response of Vehicle F under high-temperature air-conditioning conditions was jointly associated with co-emitted CO behavior, engine load and speed-related operation.

Overall, the machine-learning results show that air-conditioning operation did not

affect CO, NOx and PN through a uniform mechanism. For Vehicle B's CO, the dominant sensitivity shifted from torque at the reference temperature to NOx-related co-emission behavior under high temperature. For Vehicle E's NOx, the high-temperature air-conditioning scenario produced a clear increase in torque contribution, indicating a stronger load-related NOx response. For Vehicle F–PN, the sensitivity structure was more complex, with contributions from CO, torque and speed varying among scenarios.

4. Conclusions

This study examined CO, NOx, and PN emissions from six heavy-duty diesel vehicles over vehicle-specific CHTC cycles at 32 and 42 °C with air conditioning off and on. The results demonstrate pollutant-specific and vehicle-specific responses to air-conditioning operation.

At the reference temperature of 32 °C, air-conditioning operation increased whole-cycle distance-based PN emission factors for all six vehicles, while CO responses were modest and NOx responses varied in direction among vehicles. Work normalization generally reduced the apparent PN increases but did not remove their positive whole-cycle response. Stage-resolved and transient analyses showed that these changes were unevenly distributed over the cycles: CO increases were mainly associated with low-speed or urban stages, while NOx and PN responses depended on the vehicle–stage combination.

High temperature did not uniformly increase the signed AC response. CO interaction indices were mostly negative, whereas Vehicle E's NOx and Vehicle F's PN showed the largest positive distance-based indices, at 123.7 and 82.7 pp, respectively. Exploratory SHAP analysis indicated scenario-dependent associations between emissions and the selected operating and co-emission variables.

Further evaluation should prioritize low-speed or urban operation with air conditioning on and high-temperature AC-on conditions, particularly the NOx response of Vehicle E and PN response of Vehicle F. These findings apply to the six tested vehicles and the investigated laboratory conditions.

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Acknowledgements

This work was supported by the National Key Research and Development Project (No. 2023YFC3705600), Open Research Fund of Key Laboratory for Vehicle Emission Control and Simulation of the Ministry of Ecology and Environment, Chinese Research Academy of Environmental Sciences (VECS2024K10) and the Fundamental Research Funds for the Central Public-interest Scientific Institution (2024YSKY-03).

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Highlights

- Air conditioning altered CO, NOx, and PN emissions from heavy-duty diesel vehicles.
- PN showed the most consistent increase under air-conditioning operation.
- Emission responses varied across driving stages and transient events.
- High temperature modified AC effects in pollutant- and vehicle-specific ways.
- Auxiliary cooling loads should be considered in hot-weather emission assessments.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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